

Assignment – Word Embedding in Natural Language Processing

In this assignment, we will follow the instructions in the following Google Colab Notebook:

[Word2vec.ipynb](#)

We will explore a large Twitter dataset related to ChatGPT and generate word vectors. Then you will explore the relationship between words using the vectors.

- (1) State a hypothesis, for example: your hypothesis can be “ChatGPT is closer to OpenAI than Tesla”. Please be Creative! Also remember all the terms should be lemmatized.
- (2) Test whether the word vectors support your hypothesis.
- (3) Take a screenshot of the above test.
- (4) Then follow the instructions at the end of the notebook to do visualization using <https://projector.tensorflow.org/>
- (5) Play around with the visualization to generate insights, and take a screenshot. Add some short interpretations of your novel insights.
- (6) Submit the above screenshots in a word file.

This is individual effort, but you are encouraged to collaborate with your team to inspire each other. Your submitted work should be unique and cannot be the same as others.